JUSTICE TRÉSOR NGOM, PhD

MOLECULAR BIOLOGIST | TB GENOMICS | MICROBIAL GENOMICS | INFECTIOUS DISEASE RESEARCH

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Stellenbosch University · Faculty of Medicine and Health Sciences · South Africa

Molecular biologist and infectious-disease researcher with a PhD in Molecular Biology from Stellenbosch University and postdoctoral experience in microbial genomics. Research expertise spans tuberculosis genomics, antimicrobial resistance, molecular epidemiology, pathogen evolution and bioinformatics. Experienced in whole-genome and targeted sequencing workflows, mycobacterial laboratory research, variant interpretation, phylogenetics and genomic surveillance. Has contributed to research involving more than 3,000 clinical Mycobacterium tuberculosis isolates and to molecular surveillance programmes for emerging and zoonotic pathogens. Interested in connecting pathogen genomic evidence with biological investigation, including emerging work on host–pathogen communication and extracellular vesicles. Experienced in collaborative research, scientific writing, peer review, teaching, mentoring and research capacity strengthening.

RESEARCH INTERESTS

Tuberculosis genomics and drug resistance; molecular epidemiology and pathogen evolution; genomic surveillance; genotype–phenotype relationships; molecular diagnostics; host–pathogen interactions; extracellular vesicles and microbial communication.

EDUCATION & ACADEMIC APPOINTMENTS

Postdoctoral Research Fellow – Microbial Genomics

2025–present

TB Genomics Research Group, Division of Molecular Biology and Human Genetics, Faculty of Medicine and Health Sciences, Stellenbosch University, South Africa.

PhD, Molecular Biology

2021–March 2025

Faculty of Medicine and Health Sciences, Stellenbosch University, South Africa. Thesis: “The impact of changes in treatment strategies on the XDR-TB population structure in the Western Cape region over time.”

MRes, Molecular and Cellular Biology

2014–2016

Faculty of Sciences, University of Douala, Cameroon.

MSc, Molecular and Cellular Biology

2013–2014

Faculty of Sciences, University of Douala, Cameroon.

BSc, Biochemistry

2009–2013

Faculty of Sciences, University of Douala, Cameroon.

RESEARCH & PROFESSIONAL EXPERIENCE

Postdoctoral Research Fellow

Mar 2025–present

TB Genomics Research Group, Division of Molecular Biology and Human Genetics, Stellenbosch University, South Africa

- Investigating how changes in diagnostic algorithms and drug-resistant TB treatment strategies have shaped the population structure, evolution and transmission dynamics of M. tuberculosis in the Western Cape using WGS and bioinformatics.
- Contributed to large-scale genomic research involving more than 3,000 clinical isolates, with a focus on the genetic basis of antimicrobial resistance.
- Optimised and applied WGS and targeted-sequencing bioinformatics workflows; performed genotyping, variant interpretation and resistance biomarker analyses.
- Contributed to genomic surveillance, collaborative translational genomics initiatives, manuscript preparation, peer review and grant-development activities.
- Provided bioinformatics support and guidance to postgraduate researchers and contributed to scientific workshops and conferences.

Doctoral Research Fellow

Feb 2021–Mar 2025

TB Genomics Research Group, Division of Molecular Biology and Human Genetics, Stellenbosch University, South Africa

- Completed doctoral research on the population structure of XDR-TB in the Western Cape over time, examining the context of changing treatment strategies.
- Led genomic analyses of large clinical isolate collections, identifying resistance-associated mutations, including uncommon variants, and exploring their potential relevance to resistance and treatment outcomes.
- Integrated molecular microbiology, WGS, bioinformatics and statistical approaches to investigate genotype–phenotype relationships and bacterial adaptation under drug pressure.
- Conducted M. tuberculosis culture, nucleic-acid extraction, PCR, electrophoresis, spoligotyping, sequencing library preparation and WGS in appropriate BSL-2/3 environments.
- Applied R, Python and Bash/Linux for genomic data processing, statistical analysis and interpretation; contributed to biosafety, first-aid and fire-safety activities.

Research Laboratory Scientist / Consultant

Mar 2017–Apr 2021

Metabiota Cameroon and Centre for Army Health Research (CRESAR), Yaoundé, Cameroon

- Worked on USAID Emerging Threats Program PREDICT-2 and molecular COVID-19 diagnostic activities, supporting molecular surveillance of zoonotic and emerging pathogens.
- Performed nucleic-acid extraction, cDNA synthesis, RT-PCR, real-time PCR, nested PCR, Sanger sequencing preparation, electrophoresis, gel purification and cloning.
- Contributed to 56,601 molecular tests across human and animal surveillance activities and supported detection of 143 viruses (118 predicted and 25 previously known, as reported in project records).
- Designed and evaluated primers, supported assay optimisation and validation, and contributed to field laboratory quality assurance and biosafety.

Consultant Laboratory Technician

May–Sep 2020

Labyrinth Global Health, Inc., United States / Cameroon

- Supported standardised molecular COVID-19 diagnostic workflows and trained CRESAR personnel in sampling, transport logistics, cold-chain management, antigen testing and real-time PCR.
- Developed and implemented SOPs and supported quality assurance, biosafety, equipment calibration, troubleshooting and laboratory documentation.

Research Laboratory Scientist (Intern)

Oct–Dec 2016

Molecular Biology Laboratory, International Reference Centre Chantal Biya, Yaoundé, Cameroon

- Performed DNA/RNA extraction, RT-PCR, nested PCR, cDNA synthesis and electrophoresis; contributed to HIV molecular genotyping and drug-resistance testing using Sanger sequencing.

Graduate Research Student (MRes)

Jun–Dec 2015

National Reference Laboratory of Mycobacteriology, Centre Pasteur du Cameroun, Yaoundé, Cameroon

- Conducted research on molecular characterisation of extrapulmonary TB strains in Cameroon (2006–2015).
- Performed mycobacterial culture, microscopy and staining, biochemical testing, molecular diagnostics, GeneXpert, line-probe assays, spoligotyping, MIRU and LSP analysis.

Research Laboratory Scientist (Intern)

Aug–Nov 2014

USAID Emerging Threats Program PREDICT-1 / Global Viral Forecasting Initiative, Yaoundé, Cameroon

- Supported molecular surveillance of viral zoonoses through nucleic-acid extraction, real-time PCR, RT-PCR, nested PCR and preparation of samples for Sanger sequencing.

CORE TECHNICAL EXPERTISE

Tuberculosis & infectious diseases: Drug-resistant TB; mycobacterial diagnostics; molecular epidemiology; pathogen surveillance; zoonotic and emerging infections; genotype–phenotype analysis.

Genomics & bioinformatics: Whole-genome sequencing; targeted sequencing; variant calling and interpretation; SNP analysis; population genomics; phylogenetics and phylogenomics; transmission analysis; genome annotation.

Software & computational methods: R; Python; Bash/Linux; TBProfiler; MTBseq; Snippy/Snippy-core; IQ-TREE; TreeTime; BEAST; MAGMA; Stata; Conda; Nextflow; Snakemake.

Molecular & microbiology methods: DNA/RNA extraction; cDNA synthesis; PCR, RT-PCR and qPCR; primer design; Sanger sequencing; NGS library preparation; mycobacterial culture; GeneXpert; line-probe assays; spoligotyping; MIRU; microscopy and staining.

Laboratory quality & safety: BSL-2/BSL-3 laboratory practices; SOP development; sample and metadata documentation; assay validation; quality assurance; decontamination and waste management.

Research & leadership: Experimental design; statistical and epidemiological analysis; scientific writing; grant development; literature review; project coordination; peer review; teaching; mentoring and training.

TEACHING & TRAINING EXPERIENCE

Trainer Assistant – H3ABioNet Next Generation Sequencing Bioinformatics Course — **Aug 2023–Sep 2026**

- Supported participants with introductory Bash, Linux command-line concepts and practical bioinformatics exercises.

Part-time Lecturer Assistant, Biochemistry Department, University of Douala — **Jan 2017–Apr 2021**

- Supported teaching in bioinformatics and molecular biology, prepared and graded assessments, and organised laboratory practical sessions.

Part-time Lecturer, High Professional Institute of Health Science, Sangmelima — **Feb 2019–Feb 2021**

- Lectured in molecular biology and supported examinations and laboratory practicals.

Part-time Lecturer, High Institute MBASSI, Monatele — **Feb 2019–Feb 2021**

- Lectured in molecular biology and supported assessment and practical teaching.

Facilitator / Organiser, USAID PREDICT–Metabiota Cameroon workshop — **Aug 2019**

- Helped organise and deliver practical training in nucleic-acid extraction, cDNA synthesis, PCR and sequence analysis for Master's and PhD students.

MENTORING & SUPERVISION

MSc co-supervision, Stellenbosch University — **2025–2026**

- Genomic characterisation and evolution of drug-resistant M. tuberculosis in patients with persistent culture positivity in the Western Cape, 2019–2023.

PhD student mentoring, Stellenbosch University — **2023–2026**

- Omics approaches for discovery of diagnostic biomarkers for tuberculous meningitis.

MSc-to-PhD mentoring, University of Douala — **2018–2025**

- Molecular typing and antibiotic-resistance profiles of enteric Escherichia coli pathovars in Douala, Cameroon.

MSc-to-PhD mentoring, University of Douala — **2019–2026**

- Detection of influenza viruses and molecular characterisation of circulating strains in Cameroon, 2019–2023.

SELECTED PEER-REVIEWED PUBLICATIONS

Selected publications (full and updated publication record: Google Scholar and Stellenbosch University Researcher Profiles).

1. Mogashoa T, Ngom JT, Choga WT, et al. Whole-genome sequencing suggests recent transmission of rifampicin-resistant tuberculosis in Botswana. BMC Infectious Diseases. 2026. doi:10.1186/s12879-026-13137-5.

2. Mogashoa T, Ngom JT, Choga OT, et al. High treatment success among individuals with rifampicin-resistant tuberculosis in Botswana: A retrospective cohort study. Journal of Infection and Public Health. 2026. doi:10.1016/j.jiph.2026.103169.

3. Thagulisi F, Baatjies L, Sharma A, Ngom JT, et al. Antimycobacterial activity of intertidal sediment-derived bacteria from False Bay, South Africa. Frontiers in Microbiology. 2026;16:1745248. doi:10.3389/fmicb.2025.1745248.

4. Mogashoa T, Ngom JT, Loubser J, et al. Undetected rifampicin-resistant tuberculosis associated with rpoB I491F and V170F mutations in Botswana: Diagnostic implications. Journal of Global Antimicrobial Resistance. 2025;46:171–174. doi:10.1016/j.jgar.2025.12.005.

5. Conceição EC, Wells F, Sharma A, et al., including Ngom JT. Mycobacterium tuberculosis cultured in MGIT media for whole-genome sequencing application: a systematic literature review and meta-analysis. Microbial Genomics. 2025;11(11):001565. doi:10.1099/mgen.0.001565.

6. Ngom JT, Loubser J, Maasdorp E, et al. Population structure and emergence of resistance to new and repurposed drugs in XDR-TB: insights from a 10-year genomic study in the Western Cape, South Africa. Frontiers in Cellular and Infection Microbiology. 2025;15:1638577. doi:10.3389/fcimb.2025.1638577.

7. Mogashoa T, Loubser J, Choga OT, Ngom JT, et al. Whole-genome analysis uncovers high genetic diversity of rifampicin-resistant Mycobacterium tuberculosis strains in Botswana. Frontiers in Microbiology. 2025;16:1535160. doi:10.3389/fmicb.2025.1535160.

8. Omar S, Whitfield MG, Nolan MB, Ngom JT, Ismail N, Warren RM, Klopper M. Bedaquiline for treatment of non-tuberculous mycobacteria (NTM): a systematic review and meta-analysis. Journal of Antimicrobial Chemotherapy. 2023. doi:10.1093/jac/dkad372.

MANUSCRIPTS UNDER PEER REVIEW

1. Ngom JT, Koro Koro F, Kamdem Simo Y, et al. Drug resistance and genetic structure of Mycobacterium tuberculosis complex strains associated with extrapulmonary tuberculosis in Cameroon, 2006–2015. PLOS ONE. Under peer review (2026).

2. Pheeha SM, Ngom JT, Sharma A, et al. Examining the composition of gut microbiota in a South African population: a comparative study between type 2 diabetes mellitus patients and non-diabetic individuals. Scientific Reports. Under peer review (2026).

3. Dlamini TC, Mann B, Loubser J, Ngom JT, et al. Genomic analysis of bedaquiline resistance-associated mutations in drug-resistant Mycobacterium tuberculosis isolates and their relationship with treatment outcomes in Eswatini. BMC Medical Genomics. Under peer review (2026).

4. Zitha A, de Araujo L, Fernando T, Ngom JT, et al. Comparison of tNGS with the current drug-resistant tuberculosis diagnostic algorithm utilised in Mozambique reveals missed resistance and diagnostic escape mutations. Journal of Infection. Under peer review (2026).

5. Nono VN, Nantia AE, Mutshembele A, Ngom JT, et al. Whole-genome sequencing reveals non-canonical isoniazid resistance-associated variants in Mycobacterium tuberculosis from Cameroon. International Journal of Microbiology. Under peer review (2026).

SELECTED CONFERENCE PRESENTATIONS

1. Ngom JT, Loubser J, Kashim-Bello Z, et al. Phylogenetic analysis of extensively drug-resistant tuberculosis strains circulating in Western Cape Province, 2010–2019. ISCB-Africa / ASBCB Conference on Bioinformatics, Cape Town, South Africa, 14–17 April 2025.

2. Ngom JT, Loubser H, Klopper M, Warren R, Streicher E. Molecular epidemiology of XDR-TB in Western Cape Province, 2010–2019. Ukwanda Centre for Rural Health Research Day, Stellenbosch University, 16 May 2024. Oral presentation.

3. Ngom JT, Streicher E, Klopper M, Loubser H, Warren R. Application of WGS to predict resistance in M. tuberculosis strains to drugs in the new XDR-TB regimen. Annual Academic Day, Stellenbosch University, 31 August–1 September 2022. Oral presentation.

4. Ngom JT, Koro F, Godwe C, et al. Genotypic diversity and drug susceptibility patterns among M. tuberculosis complex isolates responsible for extrapulmonary tuberculosis in Cameroon, 2006–2015. 8th EDCTP International Forum, Lusaka, Zambia, 6–9 November 2016. Poster presentation.

5. Conceição EC, Wells F, Ismail M, et al., including Ngom JTN. Early positive culture: what is in a name? European Society of Mycobacteriology 43rd Annual Congress, Tirana, Albania, 25–28 June 2023. Poster presentation.

SELECTED PROFESSIONAL TRAINING

2025 AREF Francophone Grant Writing Programme

2025 PHA4GE Conference pre-workshop assistant

2025 International Spring School in Global Health: Host–Pathogen Interactions in Infectious Diseases

2024 Wellcome Connecting Science Global Training Course: Molecular and Genomic Approaches to Clinical Microbiology in Africa (course assistant)

2022 Wellcome Connecting Science: Next-Generation Sequencing and NGS Bioinformatics

2022 Advanced Biosafety Level 3 (BSL-3) training

2021–2022 Introduction to Python; Linux, Bash and R for biologists; bacterial genome and AMR bioinformatics courses

2019 ICH Good Clinical and Laboratory Practice E6 (R2); genomic research ethics and data-sharing practices

2019 Yaoundé Advanced Course in Immunology; advanced training in viral haemorrhagic fever diagnostics

AWARDS & FELLOWSHIPS

2025–2026 Rector's Postdoctoral Fellowship Programme, Stellenbosch University

2020 TWAS Fellowship for Research and Advanced Training

2021–2024 DAAD In-Region SUN MBHG PhD Scholarship

2022–2023 Postgraduate Scholarship Programme (PSP)

2024 Google Cloud Research Credits Programme

2020 International Fair for Young African Researchers (IFYAR) Award of Excellence

2012–2015 Presidential Awards for Excellence (undergraduate and postgraduate studies)

PROFESSIONAL AFFILIATIONS

- The VALIDATE Network – Vaccine development for complex intracellular neglected pathogens (2017–present).
- Public Health Alliance for Genomic Epidemiology (PHA4GE) (2024–present).
- International Union Against Tuberculosis and Lung Disease (2021–present).
- Initiative to Strengthen Health Research Capacity in Africa (ISHReCA) (2019–present).
- International Society for Infectious Diseases (ISID) (2016–present).
- European Society of Clinical Microbiology and Infectious Diseases (ESCMID) (2016–2018).

PEER-REVIEW SERVICE

BMJ Public Health, Scientific Reports, Infection and Drug Resistance, PLOS Medicine, Frontiers in Microbiology, Journal of Global Antimicrobial Resistance, Frontiers in Tuberculosis.

LANGUAGES & PROFESSIONAL STRENGTHS

Languages: English and French. **Professional strengths:** multidisciplinary collaboration, research coordination, scientific communication, mentoring, teaching, laboratory quality systems and capacity strengthening.

PROFESSIONAL PROFILES

LinkedIn: https://www.linkedin.com/in/justicetresorngom/

Google Scholar: https://scholar.google.com/citations?user=lPZgd4sAAAAJ&hl;=en

ResearchGate: https://www.researchgate.net/profile/Justice-Ngom

GitHub: https://github.com/justicengom

ORCID / Frontiers Loop: https://loop.frontiersin.org/people/461511/overview

References available upon request.